



EGX360 - Intelligent Stock Market Analysis and Forecasting Platform

AI-Powered EGX30 Forecasting Platform

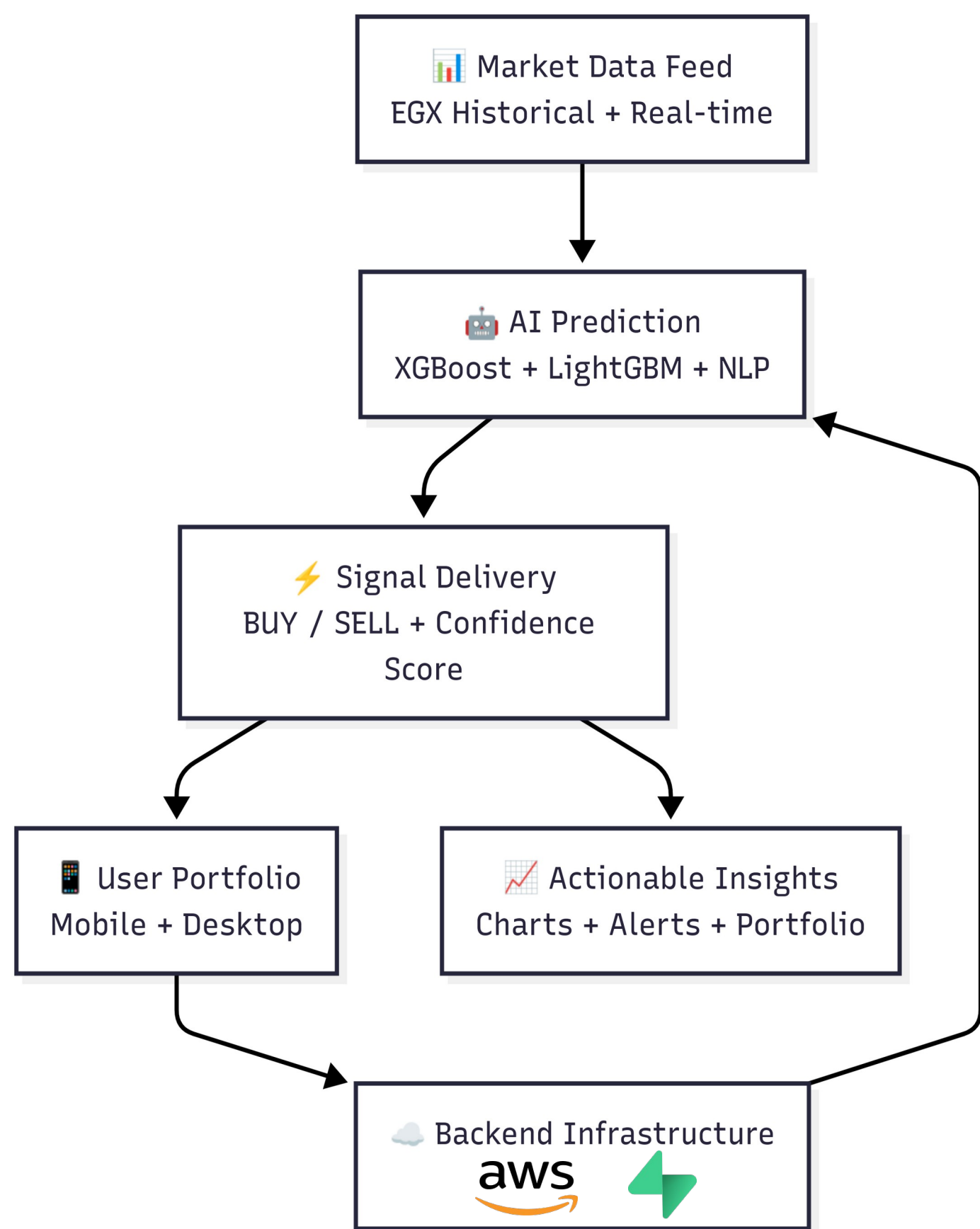
Aim& Rationale

The Problem: Egypt's retail investors make decisions with no real AI support.

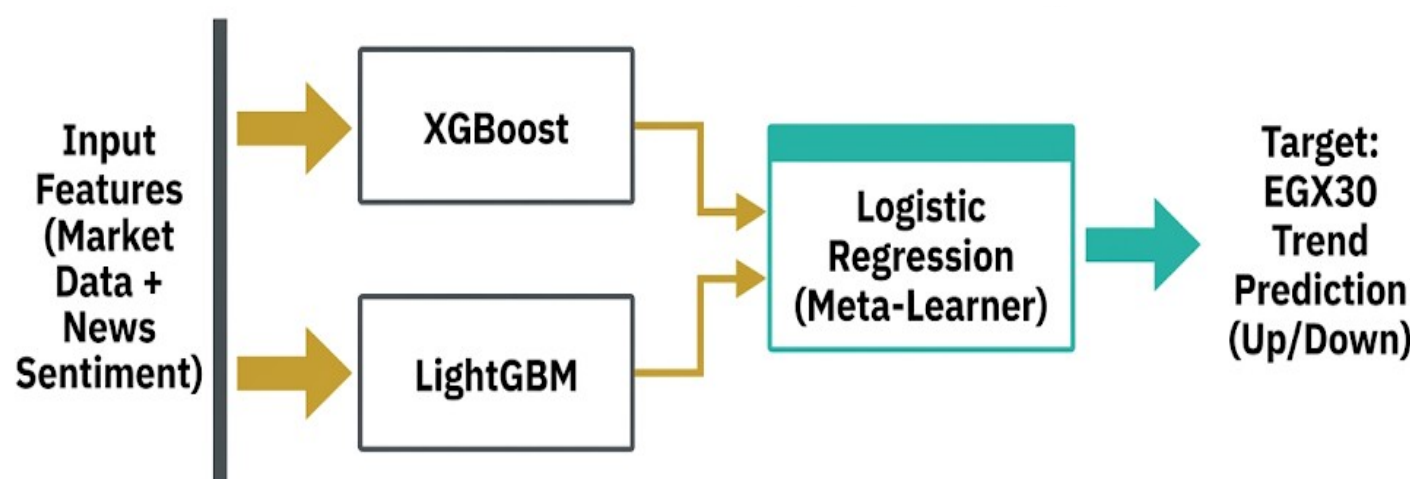
The Goal: EGX360 predicts market moves before they happen — with 90% accuracy

Rationale: Built on AWS cloud — scales to any market, any volume, in real time.

System Architecture



Stacking Ensemble



Data Processing Lifecycle

Ingestion: Daily ingestion of raw historical & news data from EGX.

Preprocessing: Optuna-optimized EMA, RSI, and volatility feature.

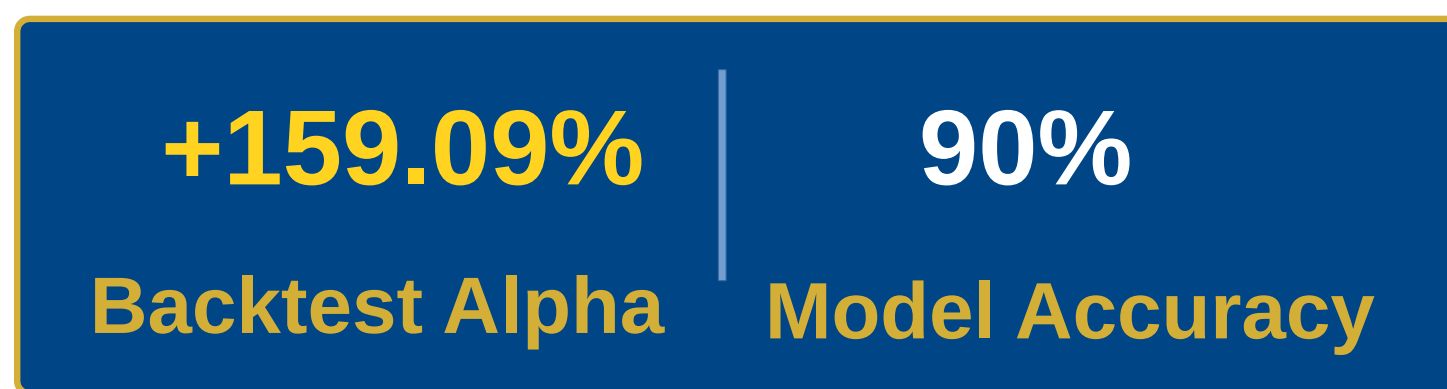
Prediction: Stacking Ensemble analysis (XGBoost + LightGBM).

Signal Delivery: Generates and delivers actionable BUY/SELL signals.

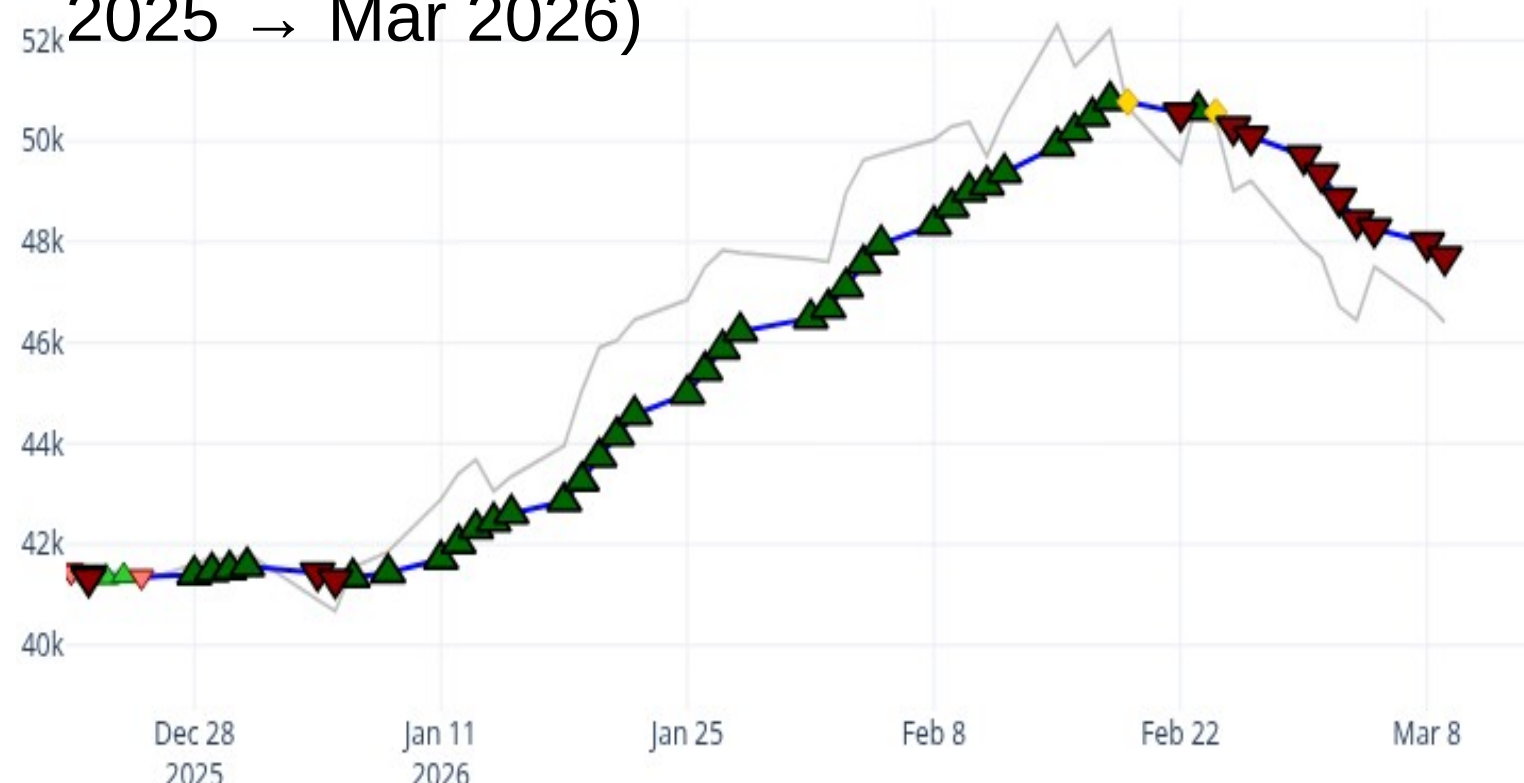
Adaptive Dashboard



Results & Performance



Live Signal Performance — EGX30 (Dec 2025 → Mar 2026)



Results & Benchmark Comparison

Study	Journal / Rank	Model Architecture	Market Context	Accuracy
Livieris et al. (2025)	Springer (Q1)	SVM, LSTM, GB	55 Global Markets	51%
Park et al. (2024)	MDPI (Q2)	LSTM vs GRU	NASDAQ	65%
Chaudhari et al. (2025)	ScienceDirect (Q2)	Ensemble + Sentiment	BSE/NSE India	72%
EGX360 (This Work)	KSIU Senior Project	Stacking (XGB + LGBM)	EGX30 (Egypt)	90%

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Lessons Learned

Technical Expertise

- **Stacking:** Optimized XGBoost & LightGBM base learners.
- **Optuna:** Bayesian optimization for robust model stability
- **Cloud:** AWS & Supabase deployment.

Core Knowledge

- **Quant:** Technical & Macro correlations.
- **NLP:** Hugging Face sentiment analysis.
- **Risk:** Automated stop-loss rules.

Problem Solving

- **Data Bias:** Strict chronological splitting
- **Noise:** EMA10 signal smoothing.

Future Work

Live Brokerage Integration:

Migrating from paper-trading to real-world execution via local brokerage APIs for automated trade execution.

Multi-Index Expansion: Adapting the Stacking Ensemble model to forecast broader indices like EGX70 and EGX100 for diversified stock

SHAP-Based Explainability:

Utilizing Explainable AI (XAI) to show users exactly how macroeconomic factors influenced each signal.

Conclusion

EGX360 achieves 90% accuracy by synergizing Egypt-specific macroeconomic indicators with a robust Stacking Ensemble. This project bridges the gap between raw data and actionable intelligence, providing retail investors with a production-ready tool for market volatility.a

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